STATIC STRUCTURAL AND FATIGUE ANALYSIS OF SUSPENSION COMPONENTS

By

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Personal Project Report



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DECLARATION

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I understand what plagiarism implies and declare that this report embodies my own ideas, words, phrases, arguments, graphics, figures, results and organization except where reference is explicitly made to another work.

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I would like to acknowledge the Department of Mechanical Engineering for providing access to the design and simulation software essential for this work. I am also thankful to my professors, colleagues, and family for their encouragement and moral support during the execution of this project and the preparation of this report.

Date: 01/09/2026

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ABSTRACT

This report presents the finite element analysis (FEA) and design evaluation of a vehicle suspension system. The suspension system is a critical subassembly responsible for absorbing road irregularities, ensuring handling stability, and transmitting loads safely. The motivation behind this study is to move beyond expensive trial-and-error physical prototyping by leveraging virtual simulation during the design phase to predict failure zones and ensure structural reliability under extreme vehicle operating conditions.

The methodology involved modelling the complete Double-wishbone Suspension assembly and generating its Bill of Materials (BOM) in SolidWorks. Key components – specifically the Upper and Lower A-Arm and Shock Absorber, were isolated, assigned real-world dimensions and engineering materials (such as Aluminium Alloy 6061-T6 and Alloy Steels), and analysed in ANSYS Workbench. For the A-Arm, a Static Structural analysis under a 10,000 N peak load and Modal analysis were conducted. For the Shock Absorber, a Static Structural analysis under a 3,000 N remote compression force and a Fatigue analysis were performed. Results demonstrated that the A-Arm experienced a maximum von Mises stress of 45.386 MPa and a total deformation of 0.0228 mm, with a minimum static factor of safety of 5.711. The Shock Absorber assembly showed a peak stress of 67.155 MPa and a maximum deformation of 7.05 mm, maintaining a safe static factor of safety of 3.85. Furthermore, durability evaluations confirmed robust fatigue lives exceeding 10^7 to 10^8 cycles for both components.

In conclusion, the finite element evaluations successfully validated the mechanical performance and safety margins of the suspension system under severe bump and cyclic loading scenarios. The primary advantage of this work lies in its capacity to optimize component designs and prevent structural failure early in development. Future scope includes integrating multi-body dynamics, terrain interaction models, and weight-optimization techniques to further enhance suspension efficiency.

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CHAPTER 1: INTRODUCTION

1.1 General Remarks

Vehicle suspension systems have been utilized for decades and have continuously evolved to become far superior to conventional fixed-axle or basic dampening setups in terms of handling and ride safety. In modern vehicle dynamics, numerous structural materials with high specific strength, exceptional fatigue resistance, and low mass are required. As lightweighting demands increase across automotive and motorsports industries (such as Formula Student, Baja SAE, and high-performance ATVs), it is essential to implement efficient design and analysis workflows that not only simplify component geometry and manufacturing but also ensure maximum structural safety under aggressive load scenarios. Leveraging virtual simulation provides a cost-effective, highly efficient mechanism to evaluate complex multi-axial forces before physical prototyping.

1.2 Introduction

The purpose of this study was to develop and formulate a comprehensive finite element analysis (FEA) model for a vehicle suspension system comprising a complete Double-Wishbone Suspension System assembly, which includes mainly the Upper and Lower A-Arm, and a Shock Absorber unit, which can then be used to evaluate structural integrity, stress distribution, deformation, and fatigue life under extreme operating conditions.

1.3 Applications

Since there are many factors/parameters that affect the water jet cutting process; the factors were listed as [1] FEM-based suspension analysis and multi-component mechanical modelling find broad applications across multiple engineering domains. Since there are many operating, geometric, and material parameters that affect the structural performance and durability of a vehicle suspension system, these key factors can be listed as:

- Vertical and cornering load magnitudes transmitted from road irregularities through the tire and hub assembly.

- Spring stiffness and pre-load characteristics of the shock absorber unit.
- Material properties (such as Young's modulus, Poisson's ratio, and yield strength) assigned to structural components like Aluminium Alloy 6061-T6 and structural steels.
- Boundary constraint configurations at the chassis mounting points and pin boss connections.
- Mesh density, element type (quadratic tetrahedral/hexahedral), and contact behaviour between mating sub-assemblies.

Building upon these parameters, applications span across:

- **Automotive Design and Passenger Cars:** Evaluating lower control arms, steering knuckles, and shock absorber mountings under maximum vertical bump and braking forces.
- **Motorsports (Formula Student / Baja SAE):** Optimizing lightweight components to meet strict structural rigidity requirements while minimizing unsprung mass.
- **Off-Road and All-Terrain Vehicles (ATVs):** Simulating extreme impact loading and high-cycle fatigue life under severe terrain conditions.

1.4 Motivation

The failure of suspension components due to structural yielding or fatigue under severe road impacts leads to catastrophic vehicle breakdown and high prototyping costs. Conventional trial-and-error physical testing is both expensive and time-consuming. Performing finite element analysis during the preliminary design stage allows for early identification of critical high-stress zones, material validation, and optimization, significantly reducing product development cycles.

1.5 Scope of Work

The scope of this term project encompasses the following steps:

- Model the complete double-wishbone front suspension assembly and generate its comprehensive Bill of Materials (BOM) in SolidWorks.
- Assign realistic engineering materials (such as Aluminium Alloy 6061-T6, ASTM A228 Music Wire, Carbon Steel 1095, and Low Alloy Steel 4140) to isolated components.

- Conduct **Static Structural** and **Modal analyses** on the lower A-Arm under a peak vertical load of 10,000 N.
- Conduct **Static Structural and Fatigue analyses** on the Shock Absorber under a 3,000 N remote compression force.
- Evaluate key metrics including total deformation, equivalent (von Mises) stress, static and fatigue factors of safety, and predicted service life cycles.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Vehicle suspension design and finite element analysis are considered core methodologies in automotive engineering, and although numerical simulations are becoming widely utilized in industry for analysing component integrity, their use is not limited to just static verification. The suspension analysis process is part of a broad engineering framework where virtual prototyping and multi-criteria optimization are implemented across the manufacturing and design sectors. The different views for this Fully Assembled Suspension System can be seen from here.

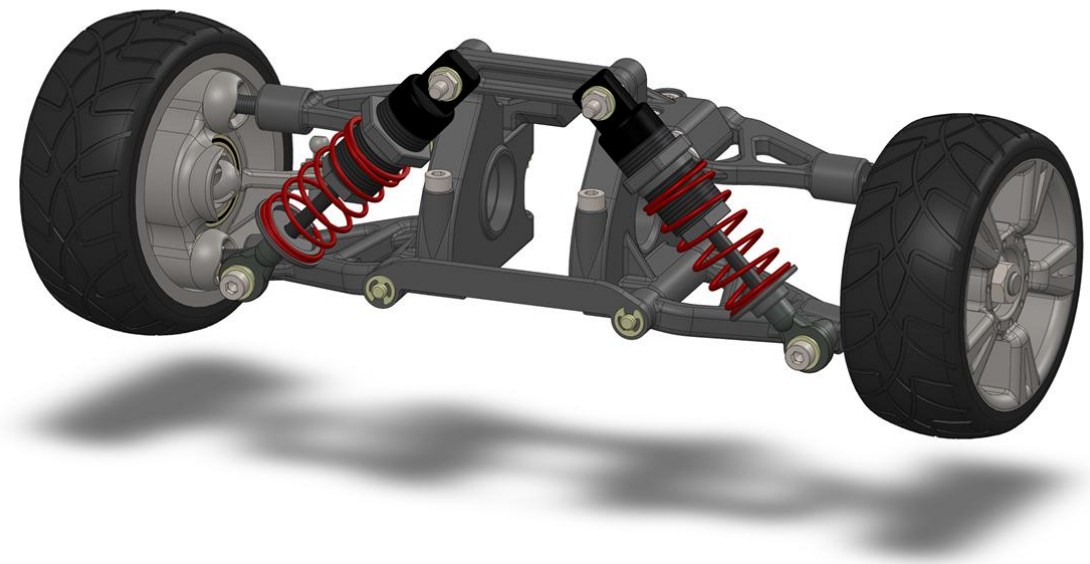


Figure 1 – Automobile Suspension System Assembly

2.2 Suspension Assembly Components

The complete front double-wishbone suspension assembly comprises numerous hardware elements, fasteners, and structural sub-assemblies as detailed in the Bill of Materials (BOM).

While every part contributes to the kinematic alignment of the system, finite element analysis (FEA) and structural validation were specifically focused on the two primary critical components that bear the maximum mechanical loads during operation:

1. **Suspension A-Arm (Control Arm):** The principal structural link connecting the wheel hub carrier to the chassis, responsible for transmitting severe lateral, longitudinal, and vertical bump loads.

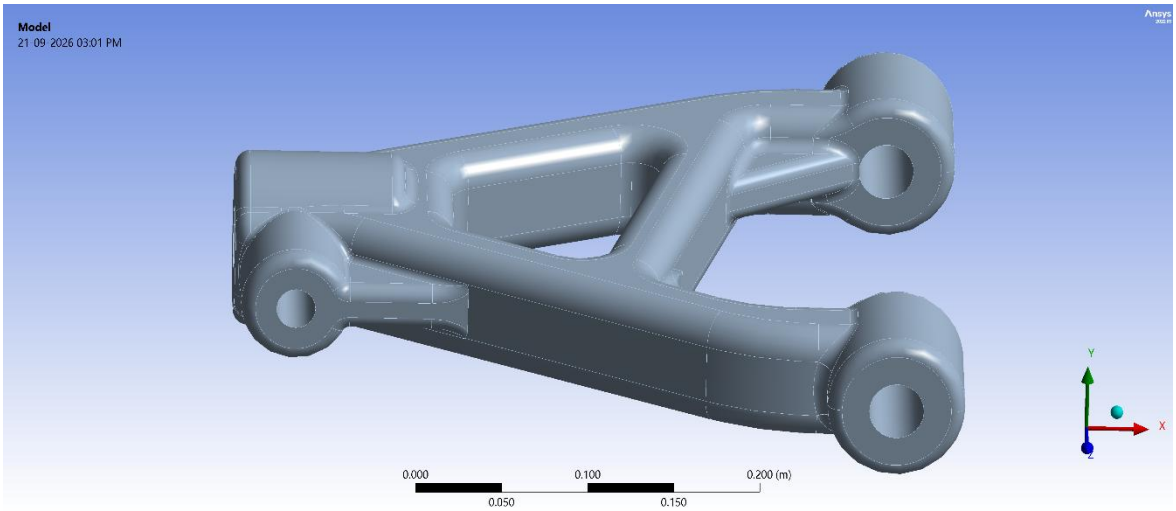


Figure 2 – Suspension Lower A-Arm (Control Arm) Model

2. **Shock Absorber Assembly:** The combined hydraulic damper and helical spring unit responsible for absorbing road impact energy and managing cyclic oscillation frequencies

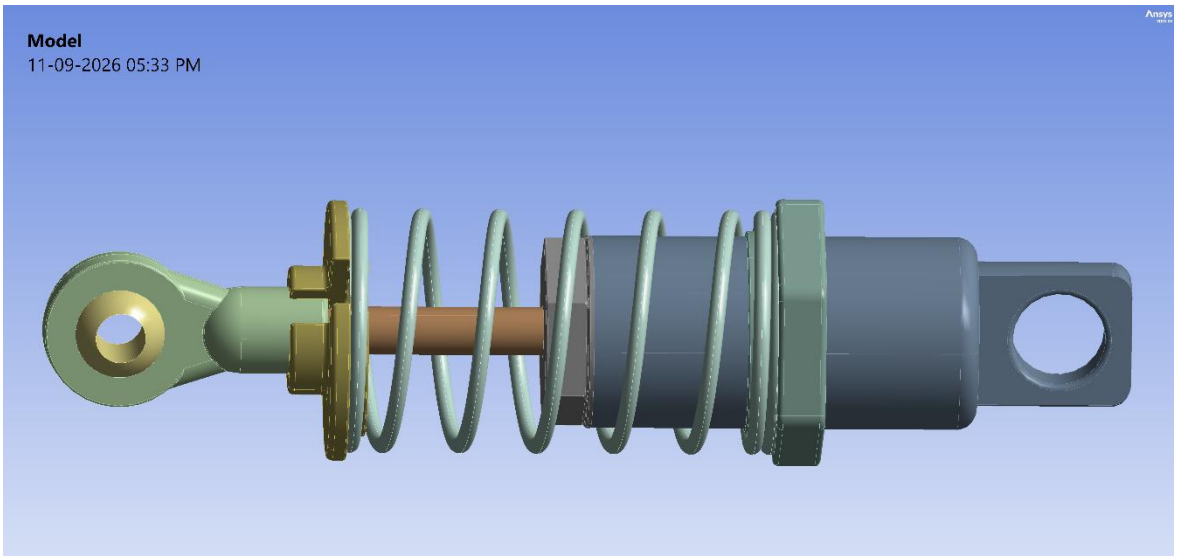


Figure 3 - Shock Absorber Assembly Model

2.3 Structural FEA of Suspension Systems

Classical analytical methods are inadequate for capturing complex geometries, stress concentrations, and multi-directional loading in suspension components. Consequently, finite element analysis (FEA) has become the industry standard for evaluating structural performance.

The equivalent (von Mises) stress criterion used to predict yielding in ductile suspension materials under multi-axial loading is formulated as given –

$$\sigma_{vm} = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}$$

where σ_{vm} is the von Mises equivalent stress, and $\sigma_1, \sigma_2, \sigma_3$ represent the principal normal stresses evaluated at critical nodes across the component geometry.

2.4 Fatigue FEA Analysis of Suspension Systems

Because suspension systems experience millions of load cycles over their operational lifespan, static structural evaluation alone is insufficient. Fatigue analysis utilizing S-N curves and cumulative damage models (such as the Palmgren-Miner linear damage rule) is necessary to predict service life and prevent high-cycle fatigue failures under repetitive cyclic loading.

2.5 Material Mechanical Properties

Table 2.1: Engineering Material Data Sources (Ansys).

Material	Density (kg/m3)	Young's Modulus (GPa)	Poisson's Ratio	Yield Strength (MPa)
Aluminium Alloy 6061-T6	2700	69.036	0.33	259.2
ASTM A228 Music Wire	7850	203	0.30	1590
Carbon Steel 1095 – Hardened and Tempered	7850	200	0.29	525
Low Alloy Steel 4140 – Hardened and Tempered	7850	205	0.29	655

CHAPTER 3: METHODOLOGY OF FEA MODELING

3.1 3D CAD Geometry and Assembly

The modelling workflow began with the development of a comprehensive 3D assembly of the vehicle's front double-wishbone suspension system using **SolidWorks**. While the baseline layout was modelled following standard automotive architecture and instructional assembly guidelines, the two primary load-bearing components—**the Suspension A-Arm and the Shock Absorber assembly**—were isolated, scaled, and updated to precise real-world physical dimensions.

3.2 Material Selection and Assignment

To accurately capture physical behaviour under operating loads, specific engineering materials were assigned to each part within the isolated assemblies:

- **Aluminium Alloy 6061-T6:** Applied to the suspension A-Arm, housing caps, and mounting eyes due to its high strength-to-weight ratio; minimizes unsprung mass to significantly improve vehicle handling and tire-road contact.

- **ASTM A228 Music Wire:** Utilized for the helical suspension spring; selected for its extreme tensile strength and elasticity, enabling repeated energy absorption under heavy impacts without permanent deformation or sag.

- **Low Alloy Steel 4140 - Hardened and Tempered:** Applied to the main piston rod and high-stress structural shafts; chosen for high fatigue strength and toughness to withstand severe axial shocks and bending moments without yielding.

- **Carbon Steel 1095 - Hardened and Tempered:** Utilized for internal locking rings and collars; chosen for high surface hardness and wear resistance to prevent joint loosening and frictional degradation under continuous vibrations.

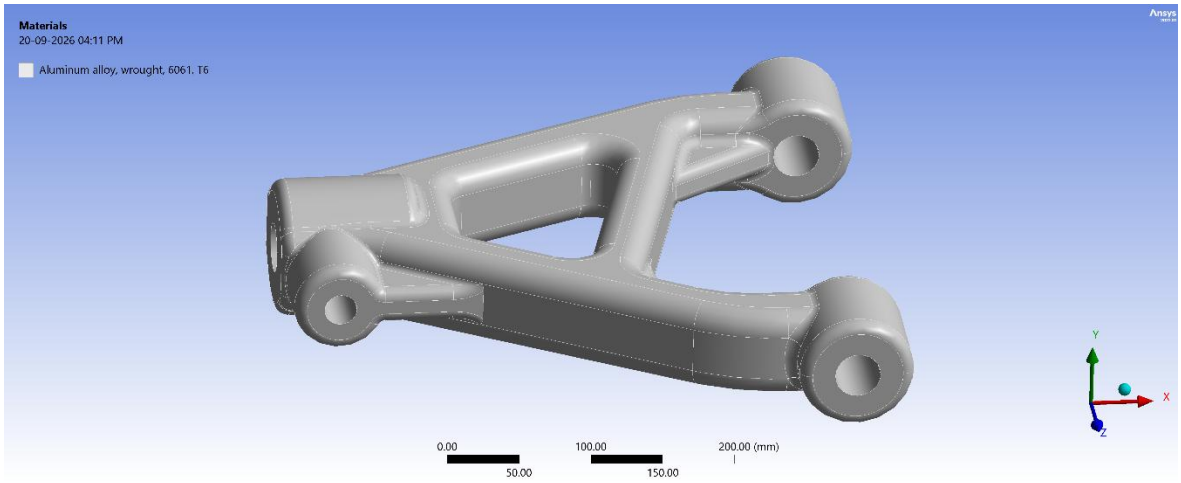


Figure 5 - Material Selection for Lower A-Arm (Control Arm)

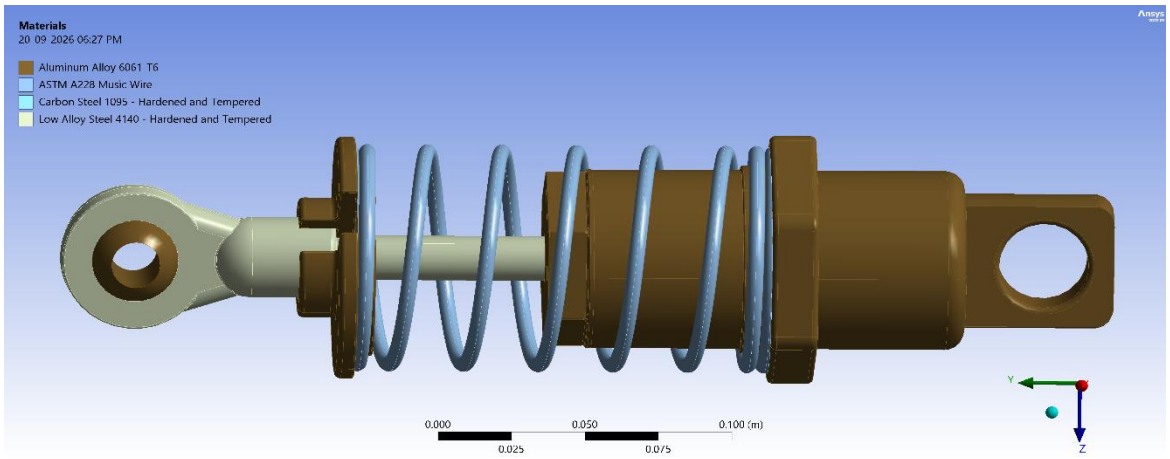


Figure 6 - Material Selection for Shock Absorber

3.3 Finite Element Discretization (Meshing)

Discretization of the continuous CAD models into finite elements was performed using ANSYS Meshing.

- An unstructured tetrahedral mesh featuring quadratic elements (higher-order elements with mid-side nodes) was generated across both components.

- Quadratic elements were explicitly chosen to handle complex curves, sharp radii, and fillet transitions effectively, ensuring accurate calculation of steep stress gradients and displacement fields.

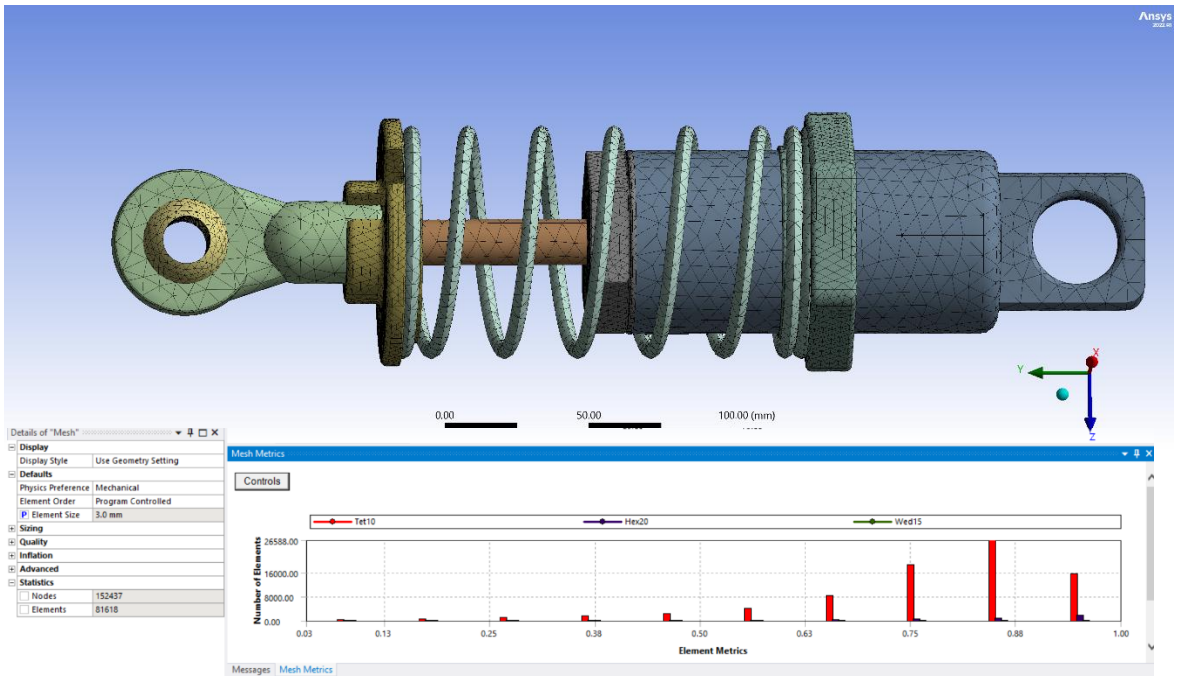


Figure 7 - Meshing Details along with Mesh Quality Metrics for Shock Absorber



Figure 8 - Meshing Details along with Mesh Quality Metrics for Lower A-Arm

3.4 Fixed Support and Loading Conditions

To replicate real-world operating environments during severe bump, cornering, and impact scenarios, precise boundary constraints and loads were applied based on an ATV / Baja SAE vehicle mass of 250 kg to 300 kg **(~2,450 N to 2,943 N total static weight)**. Divided across 4 wheels, the static weight resting on a single tire corner is roughly **612 N to 735 N**.

1. Shock Absorber Analysis Setup –

- **Fixed Supports:** A fixed support was applied to the bottom mounting eye for rigid attachment to the lower arm, and a cylindrical support (0 mm radial displacement) on the upper piston rod ensured smooth linear sliding while preventing lateral buckling.

- **Remote Force:** When the vehicle hits a severe bump, pothole, or lands from a jump, dynamic forces multiply the static corner weight (~612–735 N) by 4 times due to high g-accelerations. A **3,000 N compressive force** acts as a worst-case peak impact load to test damper structural limits and spring deflection.

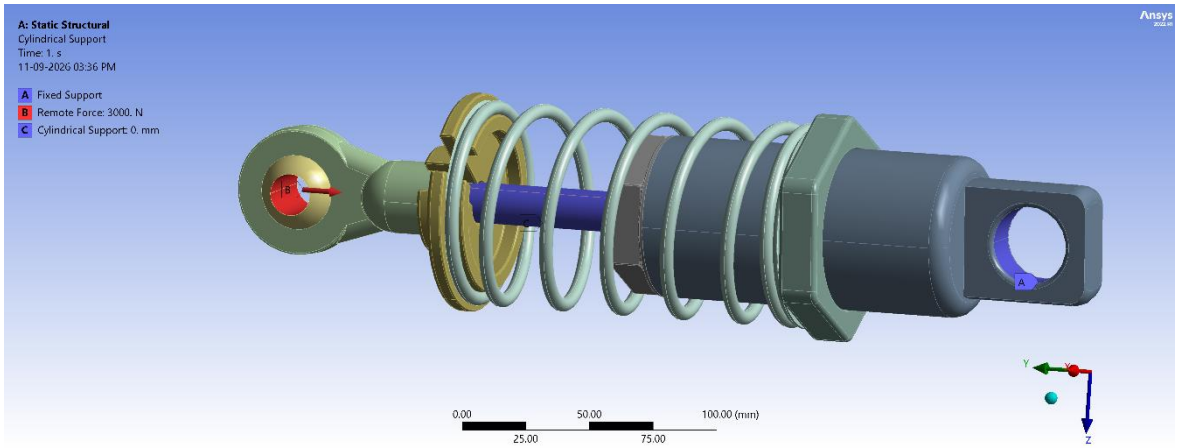


Figure 9 - Forces and Supports for Shock Absorber

2. Suspension Lower A-Arm Analysis Setup –

- **Fixed Supports:** Fixed supports were applied to the two chassis mounting hinge holes to fully constrain the pivoting control arm ends against the vehicle frame.

- **Remote Force:** The The A-Arm transmits all vertical, braking, and cornering loads from the wheel to the chassis. Starting with a static corner weight of ~612–735 N, severe 4g to 5g off-road impacts multiply this force. Multiplying this dynamic load by additional safety factors for simultaneous multi-axial stresses (braking and steering) brings the design target to a rigorous **10,000 N** upper-bound limit to verify structural yielding and fatigue safety.

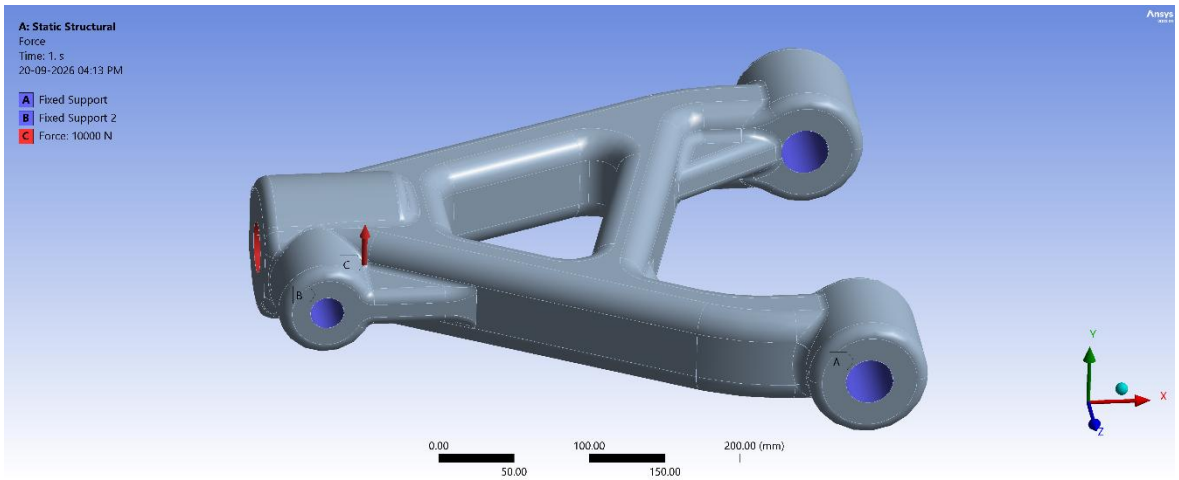


Figure 10 - Forces and Supports for Lower A-Arm

CHAPTER 4: RESULTS AND DISCUSSION

This chapter presents the finite element analysis (FEA) results for the suspension system components, specifically the shock absorber assembly and the A-Arm evaluated under the peak design loads established in the methodology.

1. Shock Absorber Analysis Results –

Given images summarize the finite element analysis outcomes for the suspension components evaluated under peak design loads:

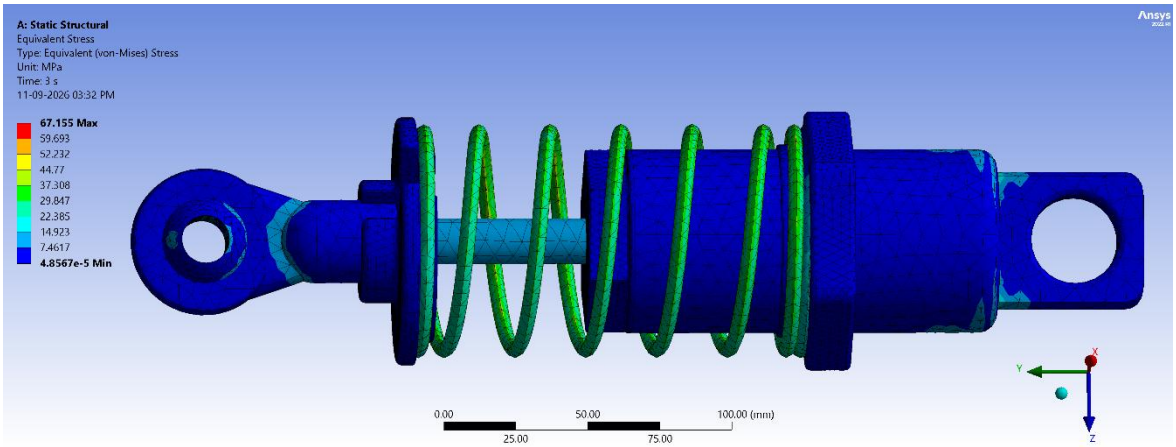


Figure 11 - Equivalent Stress on Shock Absorber

- **Discussion:** Shows a peak von Mises stress of **67.155 MPa** localized at the upper mounting eye fillet, remaining well below the material yield limit.

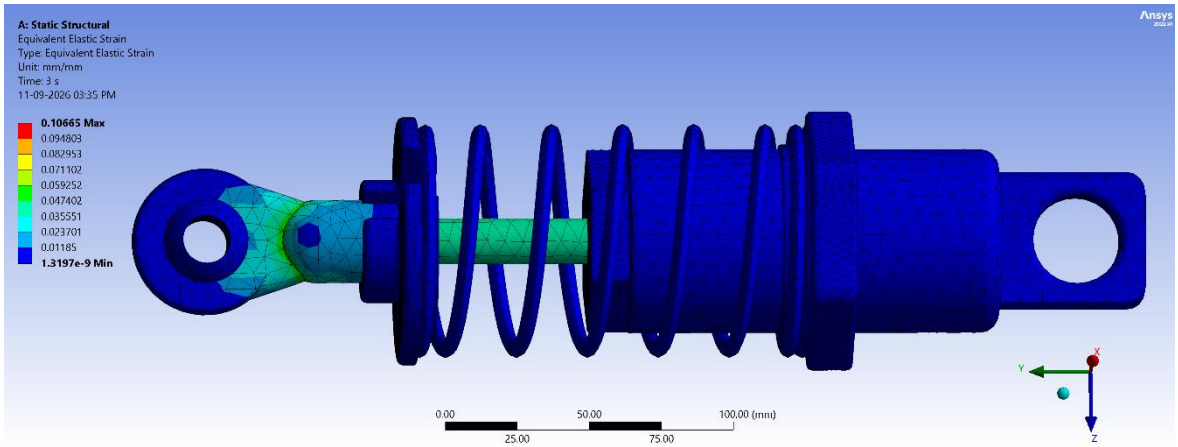


Figure 12 - Equivalent Elastic Strain on Shock Absorber

- **Discussion:** Demonstrates maximum elastic strain **0.1066 mm/mm** concentrated at the upper eye interface under compression, confirming safe elastic deformation.

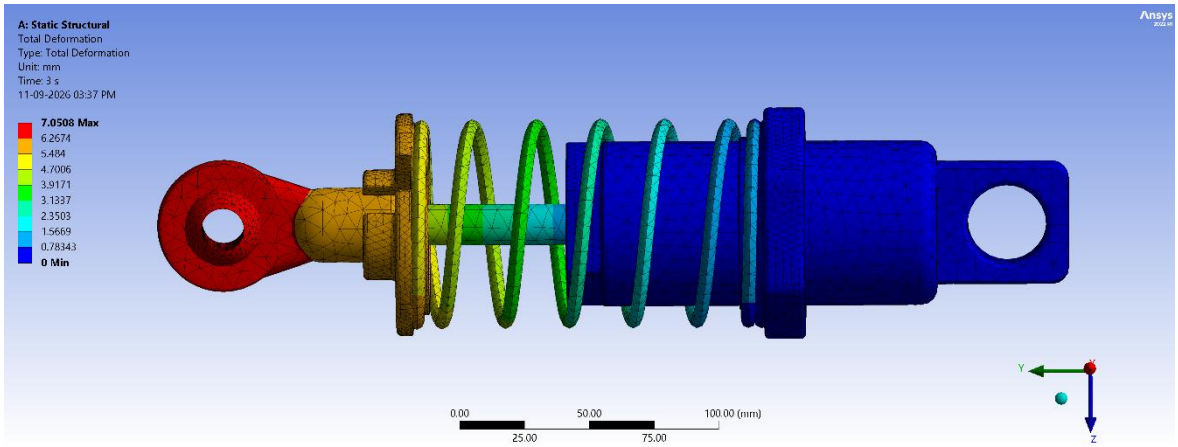


Figure 13 - Total Deformation of Shock Absorber

- **Discussion:** The maximum axial displacement peaks at **7.0508 mm** near the loaded upper mounting eye, showing smooth, controlled travel under compression

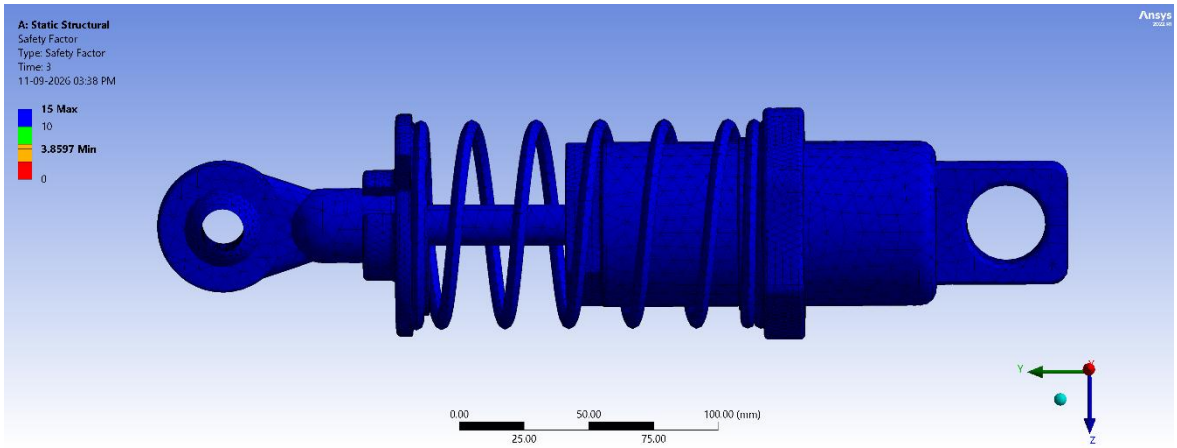


Figure 14 - Factor of Safety of Shock Absorber

- **Discussion:** The minimum static safety factor is **3.8597**, confirming a reliable margin against structural yielding under peak loading.

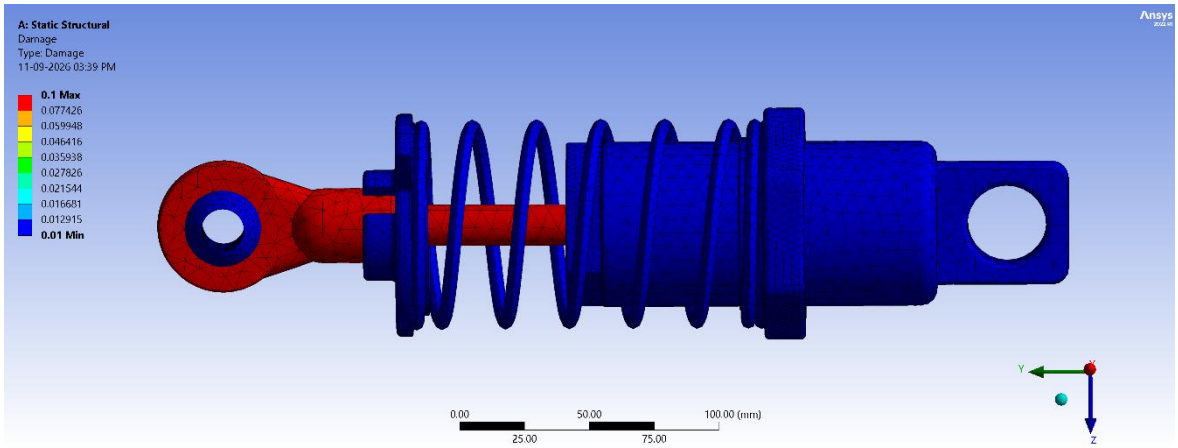


Figure 15 – Fatigue Damage to Shock Absorber

- **Discussion:** The cumulative fatigue damage remains very low, capped at a maximum index of **0.1**, proving high resilience against repetitive off-road impact cycles.

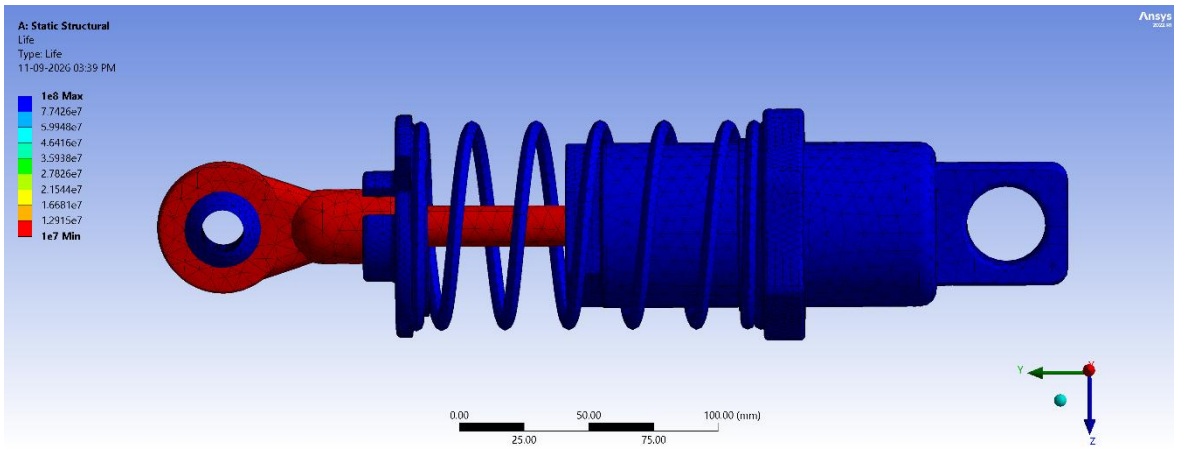


Figure 16 - Fatigue Life of Shock Absorber

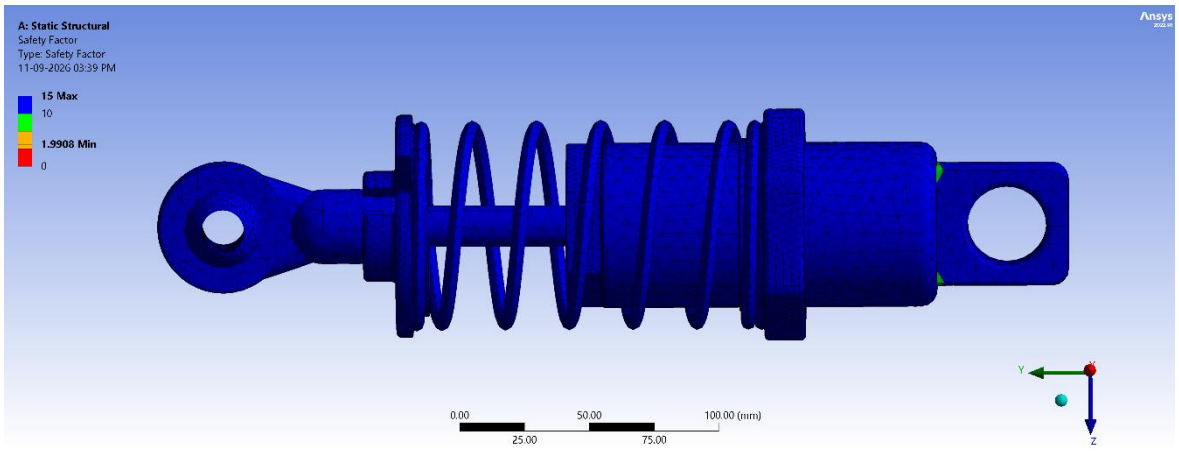


Figure 17 - Fatigue Safety Factor of Shock Absorber

- **Discussion:** Under cyclic loading, the minimum fatigue safety factor is **1.9908**, with a predicted fatigue life exceeding $\mathbf{1 \times 10^7}$ **cycles**, ensuring long-term durability.

2. Lower A-Arm (Control Arm) Analysis Results –

The simulation images summarize the finite element analysis outcomes for the suspension A-Arm evaluated under the 10,000 N peak design load.

(A) Static Structural and Fatigue Analysis:

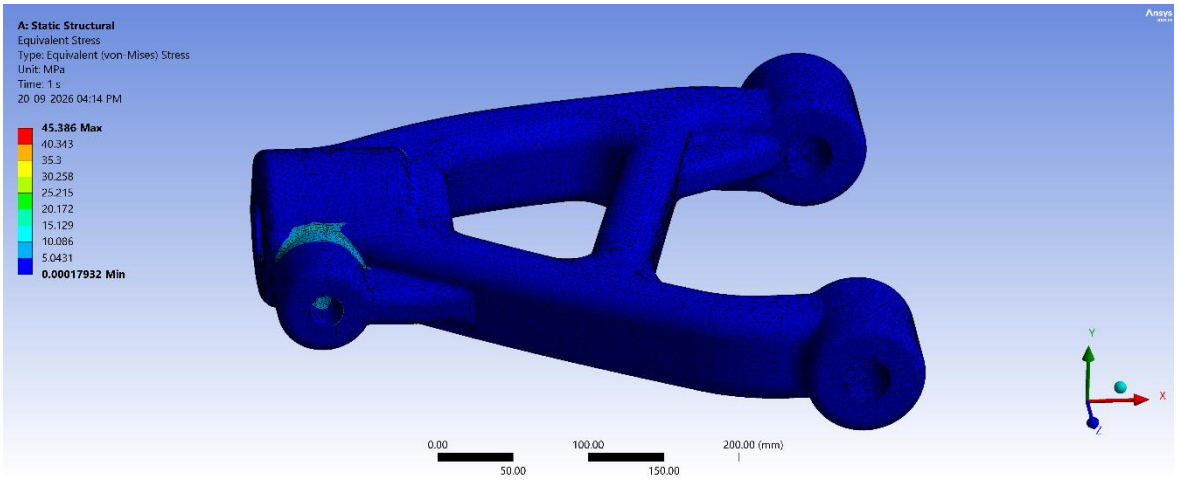


Figure 18 - Equivalent Stress on Lower A-Arm

- **Discussion:** The maximum equivalent stress peaks at **45.386 MPa**, localized specifically around the inner hinge and neck region near support B, which remains well below the yield strength of the Al 6061-T6 material.

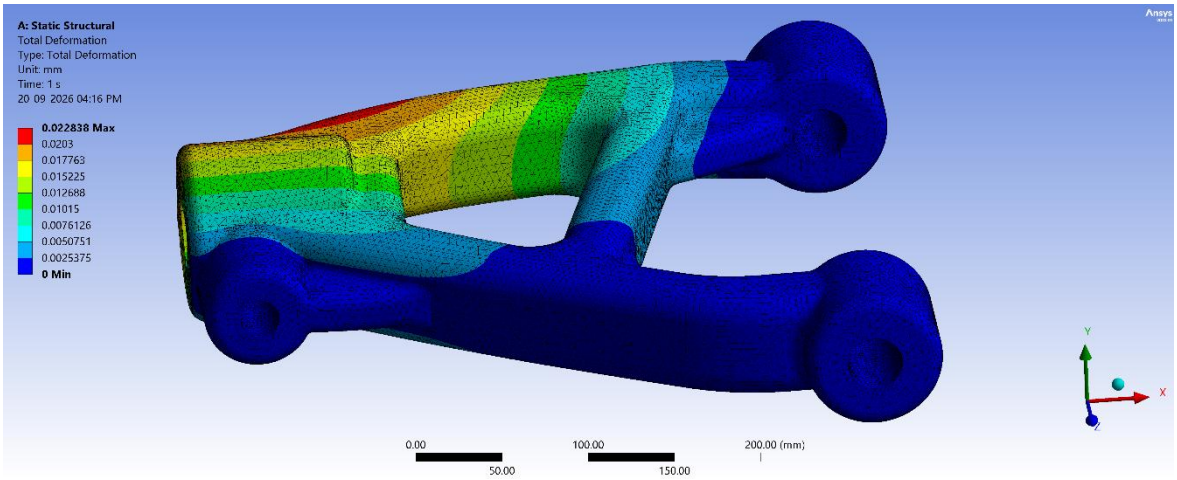


Figure 19 – Total Deformation on Lower A-Arm

- **Discussion:** Maximum structural displacement reaches **0.0228 mm** at the outer bracket section, confirming that the A-Arm maintains exceptional geometric rigidity under peak loading without excessive flexing.

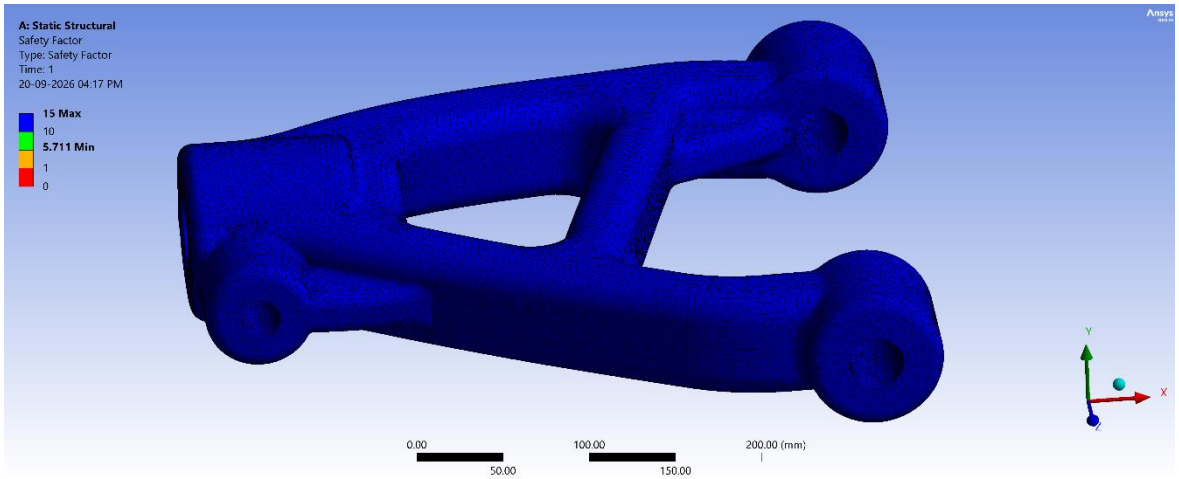


Figure 20 – Factor of Safety of Lower A-Arm

- **Discussion:** The minimum static factor of safety is **5.711**, demonstrating a robust and reliable margin against permanent plastic yielding.

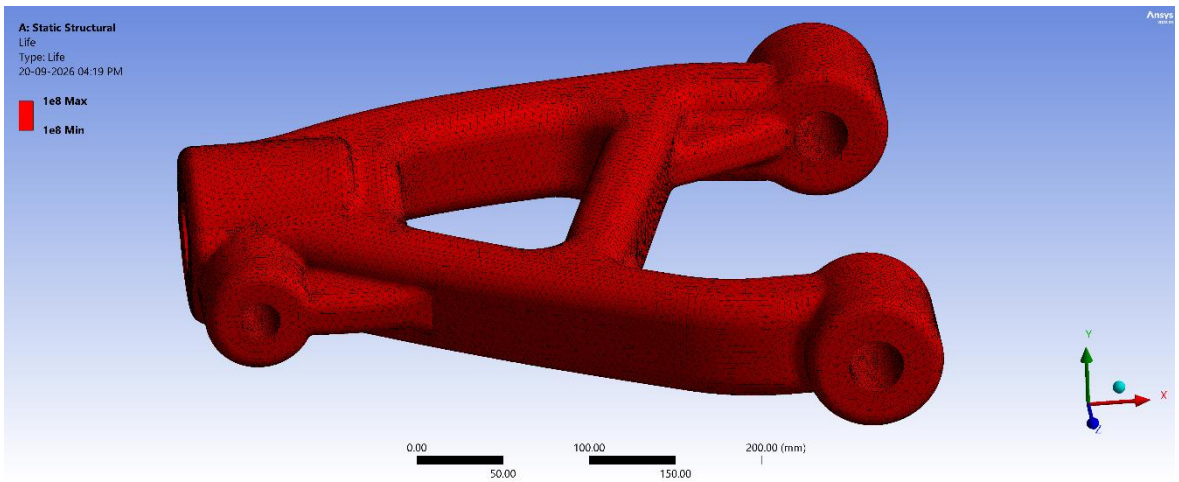


Figure 21 - Fatigue Life of Lower A-Arm

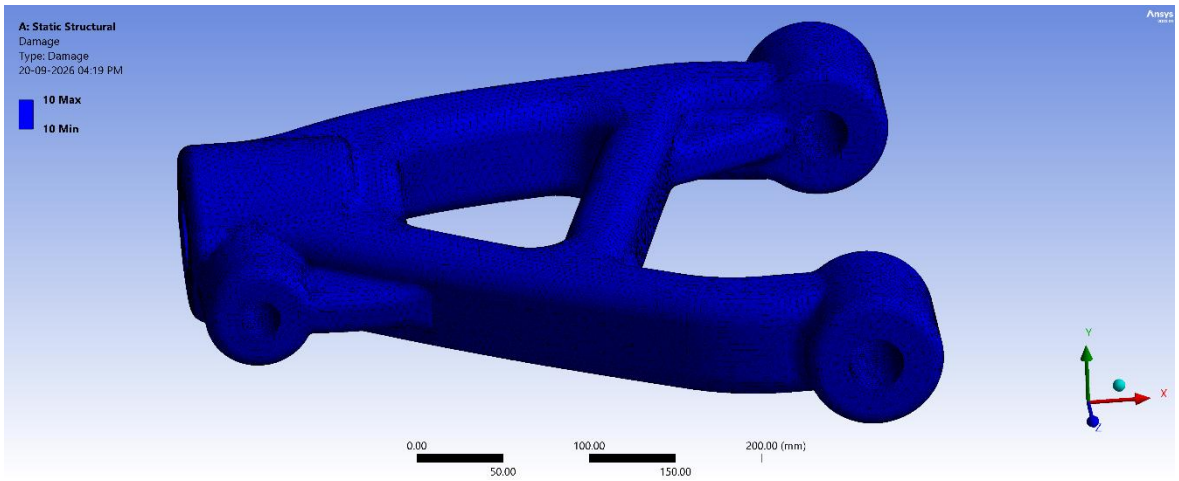


Figure 22 – Damage to Lower A-Arm

- **Discussion:** The predicted fatigue life consistently reaches the maximum threshold of 1×10^8 **cycles** across the entire body, with a negligible cumulative fatigue damage index.

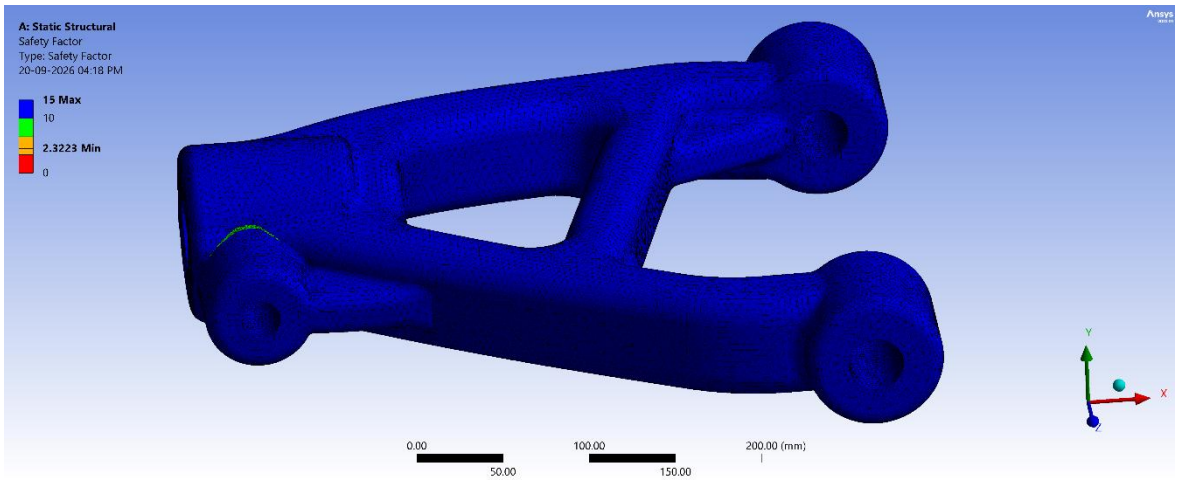


Figure 23 -Fatigue Safety Factor of Lower A-Arm

- **Discussion:** The minimum fatigue safety factor is **2.3223** located near the high-stress hinge transition, confirming outstanding endurance and resistance to cyclic failure under aggressive off-road track conditions.

(B) Modal Analysis:

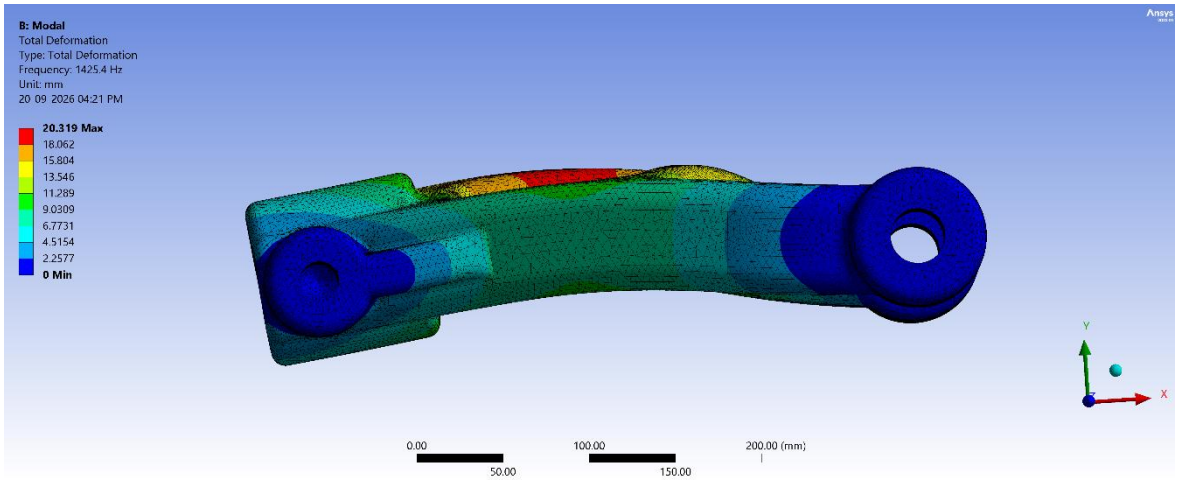


Figure 24 - Modal Shape 1 (Total Deformation 1)

- **Discussion:** Exhibits a fundamental bending mode at a frequency of **1,425.4 Hz** with a maximum relative deformation of **20.319 mm** concentrated along the central arm span.

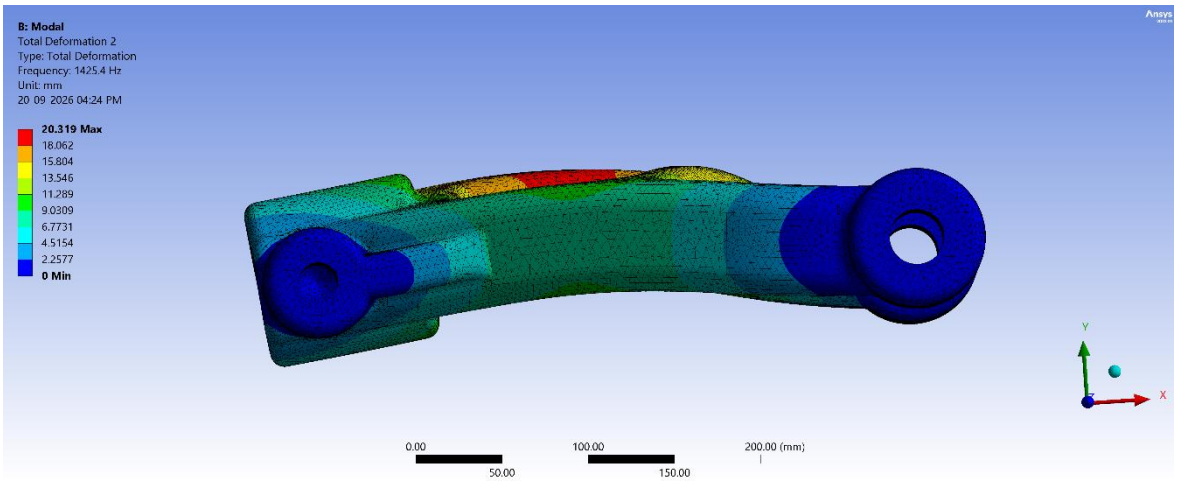


Figure 25 - Modal Shape 2 (Total Deformation 2)

- **Discussion:** Illustrates a secondary harmonic response at **1,425.4 Hz** showing identical peak relative displacement patterns (**20.319 mm**) across the main structural body.

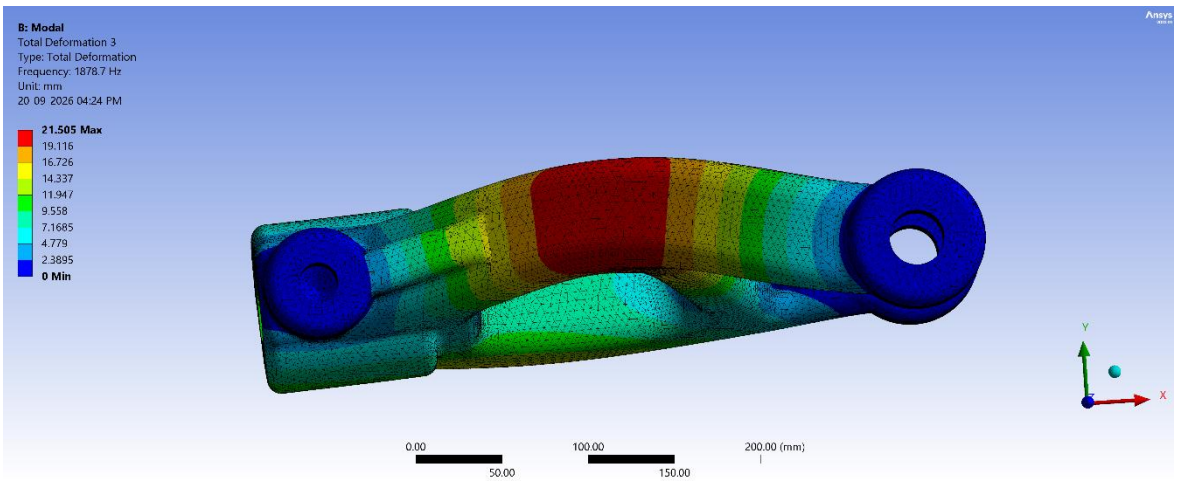


Figure 26 - Modal Shape 3 (Total Deformation 3)

- **Discussion:** Illustrates a secondary harmonic response at **1,425.4 Hz** showing identical peak relative displacement patterns (**20.319 mm**) across the main structural body.

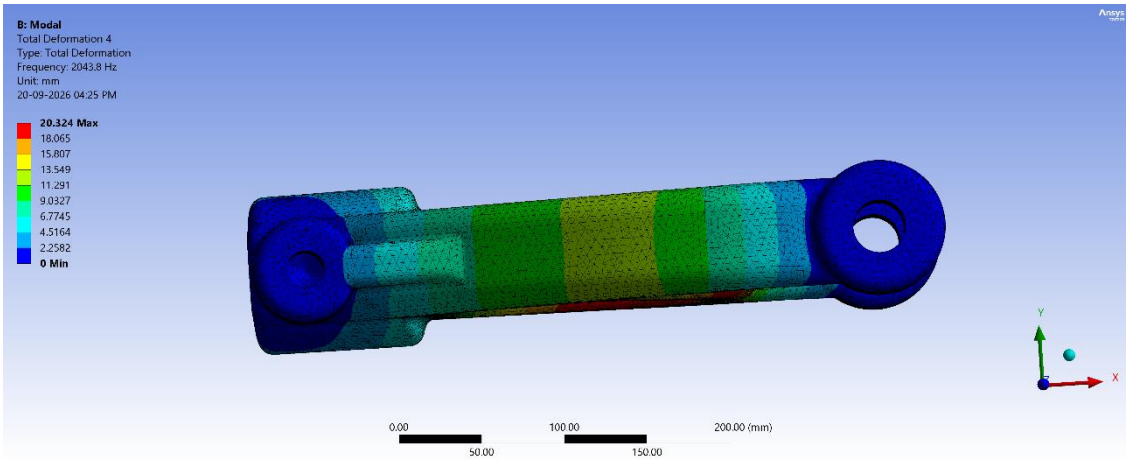


Figure 27 - Modal Shape 4 (Total Deformation 4)

- **Discussion:** Records a lateral oscillation mode at **2,043.8 Hz** reaching a peak relative displacement of **20.324 mm** along the middle shaft segment.

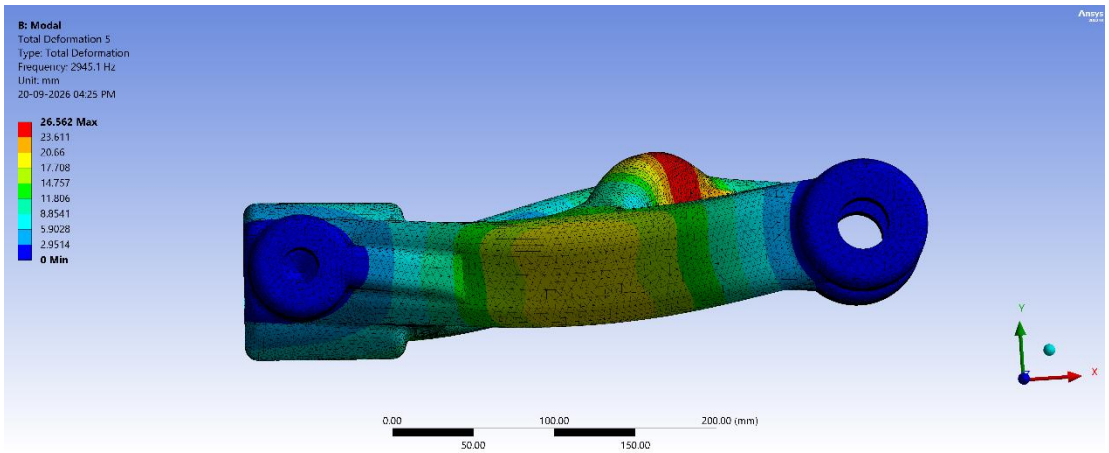


Figure 28 - Modal Shape 5 (Total Deformation 5)

- **Discussion:** Captures a higher-order out-of-plane bending mode at **2,945.1 Hz** with maximum relative deformation increasing to **26.562 mm** at the upper curved spine.

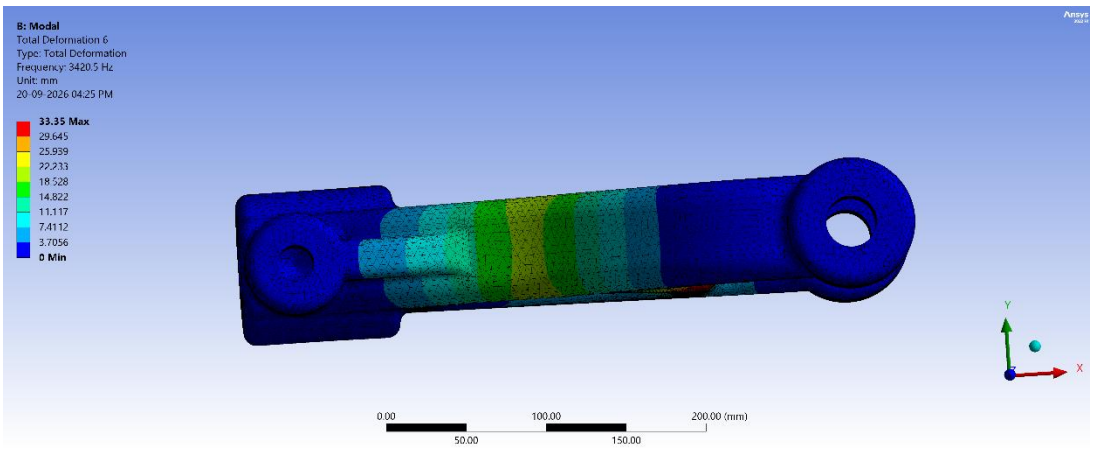


Figure 29 - Modal Shape 6 (Total Deformation 6)

- **Discussion:** Shows a complex transverse vibration mode at **3,420.5 Hz** peaking at **33.35 mm** of relative displacement along the central arm profile.

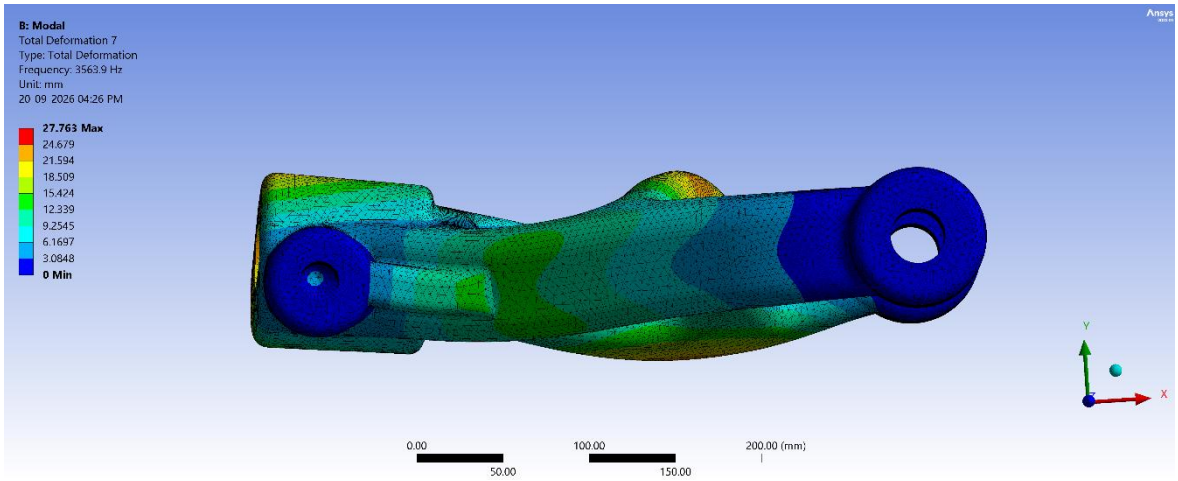


Figure 30 - Modal Shape 7 (Total Deformation 7)

- **Discussion:** Demonstrates a high-frequency multi-node modal response at **3,563.9 Hz** with a maximum relative deformation of **27.763 mm**, confirming that natural frequencies occur well above normal off-road operational input spectrums to prevent destructive resonance.

CHAPTER 5: CONCLUSION

5.1. Conclusions

The objective of this term project was to design and validate the structural integrity of the suspension system components (A-Arm and Shock Absorber) for a 250–300 kg Baja SAE/ATV vehicle under extreme off-road operating conditions.

Some of the conclusions drawn from this study include –

1. **Rigorous Load Justification:** Static corner weights (~612–735 N) were successfully scaled through $4g$-$5g$ dynamic multipliers to establish reliable upper-bound design limits of 10,000 N for the A-Arm and 3,000 N for the shock absorber.
2. **Structural Safety and Yield Limits:** Finite element analysis confirmed that all maximum von Mises stresses remain safely below the material yield strengths (reaching 45.386 MPa for the Al 6061-T6 A-Arm and 67.155 MPa for the shock assembly), ensuring all components operate completely within safe elastic limits without permanent deformation.
3. **Fatigue Endurance and Deformation:** The selected materials and geometric configurations demonstrated robust static safety factors (5.711 for the A-Arm and 3.8597 for the damper) alongside high cyclic fatigue lives (exceeding 1×10^7 to 1×10^8 cycles), proving that the design easily withstands aggressive off-road racing environments without structural failure.

5.2. Future work

In off-road vehicle suspension design and finite element analysis, individual components are studied under specific idealized load cases. Since vehicle dynamics continuously evolve with new lightweight alloy compositions and alternative composite materials being developed, there is always a need for refined multi-axial fatigue studies. Optimization

studies are also required to find the most optimum mass-to-strength operating parameter values for different structural geometries and required safety margins.

Several other studies can be conducted by expanding FEA validation to transient multi-wheel simultaneous impact scenarios, thermal effects from high-speed damper cycling, studies on maximizing component life through topology optimization, and the integration of composite or hybrid materials. And since standard structural layouts still encounter severe stress concentrations under unpredictable terrain impacts, further studies can be conducted on how to optimize joint bushings and mounting interfaces to make structural designs even more resilient for such extreme applications.

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